

Soru: $A = \begin{bmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{bmatrix}$ matrisi verilip. A matrisi-
nin özdeğerlerini ve özvektör-
lerini bulun.

Çözüm:

$$f(t) = \det(tI_3 - A) =$$

$$\begin{vmatrix} t-1 & +4 & -2 \\ 4 & t-1 & 2 \\ -2 & 2 & t+2 \end{vmatrix}$$

$$= (t-1)^2(t+2) - 16 - 16 - (4t^2 - 4t + 4t - 4 + 16t + 32)$$

$$= t^3 - 27t - 54 = (t+3)^2(t-6) = 0 \Rightarrow$$

$$(-1)^3 \cdot (-54) = 54$$

$$+1 +3 +6$$

$$t = -3 \text{ * çift kat}$$

$$t = 6$$

özdeğerler

$$f(-3) = 0$$

$$f(6) = 0$$

$t_1 = 6$ özdeğeri için,

$$\begin{bmatrix} 6-1 & 4 & -2 \\ 4 & 6-1 & 2 \\ -2 & 2 & 6+2 \end{bmatrix} \cdot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$\begin{bmatrix} 5 & 4 & -2 \\ 4 & 5 & 2 \\ -2 & 2 & 8 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 5 & 4 & -2 \\ 4 & 5 & 2 \\ -2 & 2 & 8 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \xrightarrow{\substack{S_1 \leftrightarrow S_2 + S_1 \\ S_2 \leftrightarrow S_2 + 2S_3}} \begin{bmatrix} 1 & -1 & -4 \\ 0 & 9 & 18 \\ -2 & 2 & 8 \end{bmatrix}$$

$$\begin{matrix} S_3 \rightarrow 2S_1 + S_3 \\ \rightarrow \end{matrix} \begin{bmatrix} 1 & -1 & -4 \\ 0 & 9 & 18 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{matrix} a_1 - a_2 - 4a_3 = 0 \\ 9a_2 + 18a_3 = 0 \end{matrix}$$

$$\boxed{a_2 = -2a_3}$$

$$X_1 = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 2a_3 \\ -2a_3 \\ a_3 \end{bmatrix} = a_3 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

$$a_1 = -2a_3 + 4a_3$$

$$\boxed{a_1 = 2a_3}$$

$t_{2,3} = -3$ (gitt kat) özdəyi ikih

$$\begin{bmatrix} -4 & 4 & -2 \\ 4 & -4 & 2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -4 & 4 & -2 \\ 4 & -4 & 2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \xrightarrow{\substack{S_1 \leftrightarrow S_2 \\ S_2 \leftrightarrow S_1 - 2S_2 \\ S_3 \leftrightarrow S_1 + 2S_3}} \begin{bmatrix} 4 & -4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{S_1 \rightarrow \frac{1}{4}S_1} \begin{bmatrix} 1 & -1 & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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$$a_1 - a_2 + \frac{1}{2}a_3 = 0 \Rightarrow a_1 = a_2 - \frac{1}{2}a_3$$

$$a_2 = a, \quad a_3 = b \text{ free,}$$

$$X_{23} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a - \frac{1}{2}b \\ a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix}$$

$b = 2b$ se iline,

$$X_{23} = a \underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}_{X_2} + b_1 \underbrace{\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}}_{X_3}$$

X